

TBB Cafe Website - Project Documentation

1. Project Overview

TBB Cafe Website is a full-stack cafe management and ordering platform designed to provide customers with an interactive online cafe experience and help administrators manage cafe operations.

The application allows customers to browse menus, register/login, add items to cart, place orders, view order history, and reserve tables. Administrators can manage menu items, orders, reservations, and dashboard activities.

2. Technology Stack

Frontend:

- React.js
- Vite
- React Router
- Redux Toolkit for state management
- Bootstrap and CSS for UI design
- Framer Motion for animations

Backend:

- Node.js
- Express.js
- REST APIs

Database:

- MySQL

Additional Services:

- JWT authentication
- bcrypt password encryption
- Nodemailer

3. System Modules

Customer Module:

- Customer registration
- Customer login/logout
- Browse cafe menu
- Add/remove items from cart
- Place orders
- View order history
- Table reservation

Admin Module:

- Admin authentication
- Dashboard
- Menu management
- Add menu items
- Order management
- Reservation management

4. Frontend Structure

Main components:

- Hero: Landing page section
- Navbar: Navigation system
- Menu: Displays cafe items
- MenuCard: Individual menu item display
- Cart: Shopping cart handling
- Order: Order placement flow
- Reservation: Table booking interface
- CustomerPanel: Login, registration, order history
- AdminPanel: Cafe administration features

5. Backend Structure

Backend follows a controller and route-based architecture.

Important files:

- server.js: Starts backend server
- config/db.js: Database connection
- controllers: Business logic
- routes: API endpoint definitions

APIs handle:

- Customer authentication
- Menu operations
- Orders
- Reservations

6. Authentication & Security

Customer and admin authentication are implemented using JWT tokens.

Passwords are secured using bcrypt hashing before storing them in the database.

Protected routes prevent unauthorized users from accessing restricted pages.

7. State Management

Redux Toolkit is used for global application state.

Implemented slices:

- authSlice: Authentication state
- cartSlice: Cart management
- menuSlice: Menu data handling
- orderSlice: Order information
- reservationSlice: Reservation state

8. Database Overview

MySQL database stores:

- Customer information
- Menu items
- Orders
- Reservations
- Admin related data

The backend communicates with MySQL using mysql2 package.

9. Project Workflow

Customer Flow:

Register/Login → Browse Menu → Add Items → Checkout → Place Order → View History

Reservation Flow:

Select Date/Time/Table → Submit Reservation → Admin Verification

Admin Flow:

Login → Dashboard → Manage Menu → Manage Orders → Manage Reservations

10. Future Enhancements

Possible improvements:

- Online payment integration
- Real-time order tracking
- Customer reviews
- Inventory management
- Advanced analytics dashboard
- Mobile application version

Cafe Management System - Database Structure

Database Name: cafe_db

1. Customers Table (customers)

Purpose: Stores registered customer account details and authentication information.

Column Name	Data Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique customer ID
name	VARCHAR(100)	NOT NULL	Customer name
email	VARCHAR(150)	UNIQUE, NOT NULL	Customer email
password	VARCHAR(255)	NOT NULL	Encrypted customer password
phoneno	VARCHAR (15)	NULL	Customer phone number
is_verified	BOOLEAN	DEFAULT FALSE	Email verification status
verification_token	VARCHAR(255)	NULL	Token for account verification
reset_token	VARCHAR(255)	NULL	Password reset token
reset_token_expiry	BIGINT	NULL	Reset token expiry time
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation date

2. Menu Table (menu)

Purpose: Stores all cafe food and beverage items.

Column Name	Data Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Menu item ID
name	VARCHAR(100)	NOT NULL	Food item name

Column Name	Data Type	Constraints	Description
price	DECIMAL(10,2)	NOT NULL	Item price
type	VARCHAR(50)	NOT NULL	Veg/Non-Veg/Drink
category	VARCHAR(50)	NULL	Food category
discount	INT	DEFAULT 0	Discount percentage
image	VARCHAR(500)	NULL	Food image path
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Item creation time

3. Reservations Table (reservations)

Purpose: Stores table booking details made by customers.

Column Name	Data Type	Constraints	Description
id	INT	Primary Key	Reservation ID
customer_id	INT	Foreign Key	Linked customer
name	VARCHAR	NULL	Customer name
email	VARCHAR	NULL	Customer email
date	DATE	NULL	Reservation date
time	TIME	NULL	Reservation time
guests	INT	NULL	Number of guests
phoneno	VARCHAR(15)	NULL	Customer phone number
table_no	INT	NULL	Reserved table number
status	VARCHAR(20)	DEFAULT 'Pending'	Booking status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Booking creation time

Relationship:

customers (1) ----- (Many) reservations

4. Orders Table (orders)

Purpose: Stores customer food orders.

Column Name	Data Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Order ID
customer_id	INT	Foreign Key	Customer who placed order
customer_name	VARCHAR(100)	NULL	Customer name
address	VARCHAR(255)	NULL	Delivery address
total_price	DECIMAL(10,2)	NULL	Total bill amount
status	VARCHAR(50)	DEFAULT Pending	Order status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Order date

Relationship:

customers (1) ----- (Many) orders

5. Order Items Table (order_items)

Purpose: Stores individual items inside each order.

Column Name	Data Type	Constraints	Description
id	INT	Primary Key	Item ID
order_id	INT	Foreign Key	Related order
item	VARCHAR(100)	NULL	Ordered food item
quantity	INT	NULL	Quantity ordered

Relationship:

orders (1) ----- (Many) order_items

Foreign Key:

order_items.order_id → orders.id

Delete behavior:

ON DELETE CASCADE

6. Feedback Table (feedback)

Purpose: Stores customer reviews and feedback.

Column Name	Data Type	Constraints	Description
id	INT	Primary Key	Feedback ID
name	VARCHAR(100)	NULL	Customer name
email	VARCHAR(100)	NULL	Customer email
phone_no	VARCHAR(15)	NULL	Customer phone
rating	INT	NULL	Rating (1-5)
message	TEXT	NULL	Customer feedback
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Feedback date

7. Addresses Table (addresses)

Purpose: Stores customer delivery addresses.

Column Name	Data Type	Constraints	Description
id	INT	Primary Key	Address ID
customer_id	INT	Foreign Key	Customer reference
full_address	TEXT	NULL	Complete address
city	VARCHAR(100)	NULL	City
pincode	VARCHAR(20)	NULL	Postal code

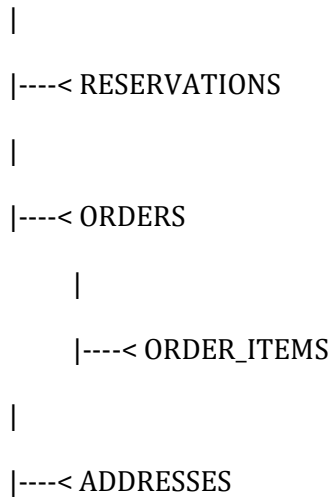
Column Name	Data Type	Constraints	Description
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Address creation date

Relationship:

customers (1) ----- (Many) addresses

Database Relationships (ER Overview)

CUSTOMERS



MENU

(Independent table used for displaying food items)

FEEDBACK

(Stores customer reviews)